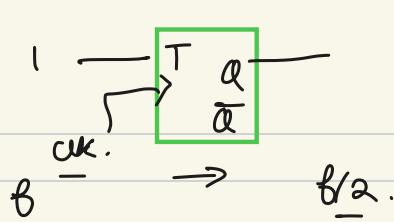
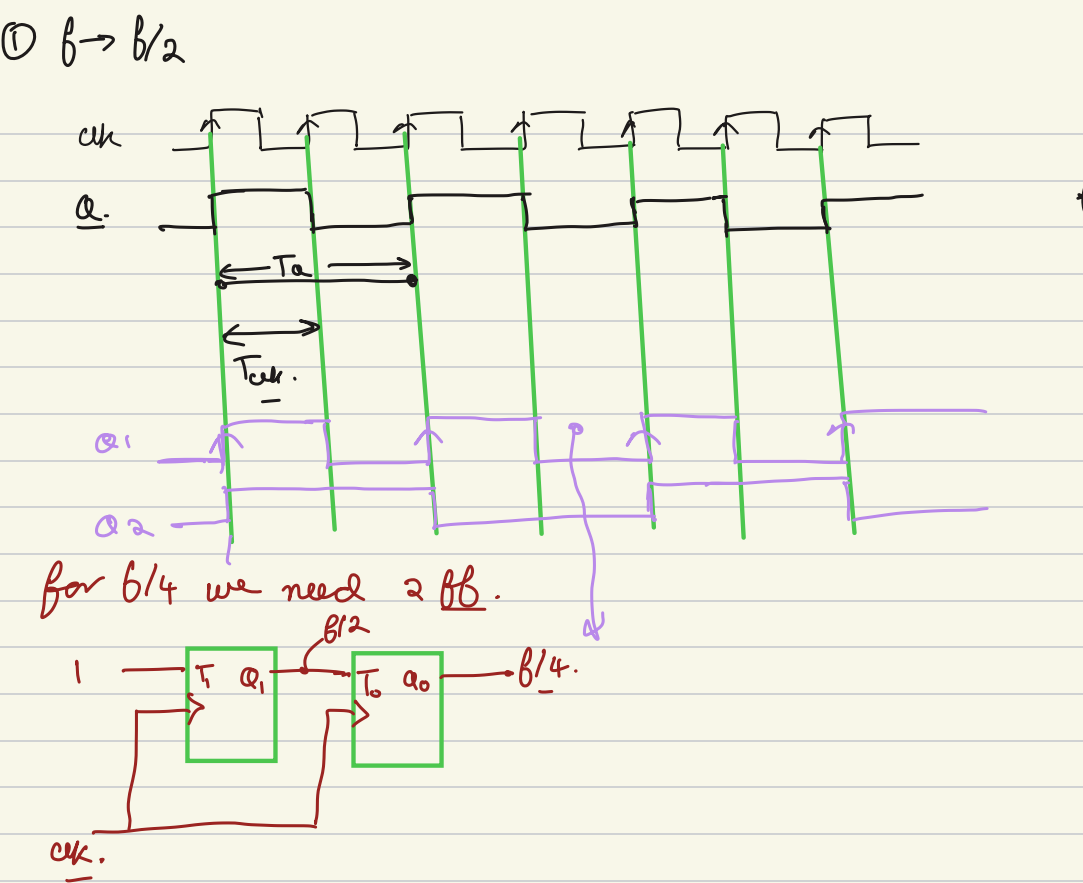


Frequency Division :

Kapil Tripathi



frequency division by

$f/2$	$f/3$	$f/1.5$
$f/4$	$f/5$	$f/2.5$
$f/8$	$f/17$	$f/3.5$

Kapil Trigarthy

In your sketch, the flip-flop is being used in **toggle mode**: a JK flip-flop does that when $J = K = 1$, and TI explicitly describes that case as a toggle flip-flop. A D flip-flop with $D = \bar{Q}$ is the same idea; university notes also show that exact connection as a **divide-by-2 counter**.

Texas Instruments +1

1) Why it is called clock division

A clock divider is named from the **signal applied to the clock terminal**. National Instruments defines frequency division as generating “a signal with a frequency that is a fraction of an input signal,” and in divider/counter circuits that “input signal” is the clock being counted. University notes describe a D flip-flop configured as a **divide-by-2 counter**, and ripple counters as a sequence of flip-flops driven by an **input clock**.

Ni +2

So in your example:

- **Clock input** = the waveform on CLK
- **T input = 1** = only a control setting that says “toggle on every active clock edge”
- **Output Q** = a new waveform whose frequency is derived from the clock

That constant 1 is **not** being divided, because a constant does not even have a frequency to divide.

2) What actually happens in a divide-by-2 flip-flop

Take a T flip-flop with $T = 1$.

At every active clock edge:

- if $Q = 0$, it becomes 1
- if $Q = 1$, it becomes 0

So the sequence is:

- after 1st clock edge: Q changes $0 \rightarrow 1$
- after 2nd clock edge: Q changes $1 \rightarrow 0$

Now Q has completed **one full cycle** only after **two clock edges**.

That is the whole reason the output frequency becomes:

$$f_{out} = \frac{f_{clk}}{2}$$



Equivalently, the period doubles:

$$T_{out} = 2T_{clk}$$

That is exactly why people call it **divide-by-2**.

3) Why it may look like “the input is being divided”

This confusion happens because flip-flops have multiple inputs:

- **clock input**
- **data/control inputs** like D, J, K, T, preset, clear

But these inputs do **different jobs**.

The **clock input** decides **when** state may change.

The **D/J/K/T input(s)** decide **how** the state changes at that clock event.

In your case, $T = 1$ does not create the frequency. It only says:

“Whenever a clock edge comes, invert the present state.”

So the **timing source** is still the clock. That is the signal being divided.

A good test is this:

- If you keep $T = 1$ but remove the clock, nothing happens.
- If you keep the clock but change the logic configuration away from toggle mode, the divide-by-2 behavior disappears.

That shows the divider is fundamentally a **clock-driven sequential circuit**.

4) Why the output becomes a slower “clock-like” signal

A divider does not physically chop a pulse in half the way a resistor divider halves voltage. Instead, it **counts clock edges** and changes state only after a pattern of those edges.

So “clock division” in digital logic means:

use sequential logic to generate a new periodic signal whose edge rate is a known fraction of the original clock.

That is why datasheets for counter/divider ICs describe them in exactly those terms. For example, Nexperia’s 8-stage divider says it is a chain of 8 flip-flops where **each flip-flop divides the previous stage by 2**, and the final output is the input frequency divided by 256. TI and Nexperia decade-counter datasheets similarly describe internal **divide-by-2** and **divide-by-5** sections for frequency division applications.

Nexperia +2

5) A very important conceptual distinction

There are two very different situations:

Case A: Clock divider

You feed a periodic signal into the **clock pin** of a sequential circuit, and the circuit generates a slower periodic output.

That is clock division.

Case B: Sampling data

You feed some changing waveform into the **D input** of a D flip-flop, while another signal clocks it.

That is **not** clock division. That is just **sampling/synchronizing/storing data**.

So if someone says “this flip-flop divides the clock,” they mean:

- the periodic signal on the **clock pin** is the reference being counted
- the output toggles at a reduced rate relative to that clock

Definition of this circuit:

This is a **clock divider** or **frequency divider**. More specifically, with a flip-flop connected in toggle mode, it is a **divide-by-2 circuit**. A frequency divider is a circuit that produces an output whose frequency is a fixed fraction of the input signal's frequency. NI defines frequency division as generating a signal whose frequency is a fraction of an input signal, and TI/Nexperia datasheets describe flip-flop divider stages exactly this way. NI +2

The thing being toggled is the flip-flop's **stored state**.

In a T flip-flop, $T=1$ only says "invert the stored bit on each clock edge."

In a D flip-flop divider, D is fed from $\overline{Q}$, so the stored bit is again inverted on each clock edge.

The constant input 1 is not what "moves through" to the output.

Duty cycle definition

Duty cycle is the ratio of the time for which a periodic signal remains in its active state to the total time period of that signal, usually expressed as a fraction or percentage. For pulse and digital waveforms, the active state is typically the **HIGH** or **ON** state. NI +1

Mathematically,

$$D = \frac{T_{ON}}{T}$$

or, in percentage form,

$$D = \frac{T_{ON}}{T} \times 100\%$$

where T_{ON} is the time the signal is active, and T is the total period. NI defines it equivalently as **High Time / Pulse Period**, with pulse period equal to high time plus low time. NI +1

In cleaner engineering language for exams:

Duty cycle is the percentage of one complete cycle during which a signal, pulse, or waveform stays at logic high or ON level. This is the standard meaning used for PWM, clock signals, and pulse waveforms. NI +1

Example:

If a signal is ON for 2 ms in a total period of 10 ms, then

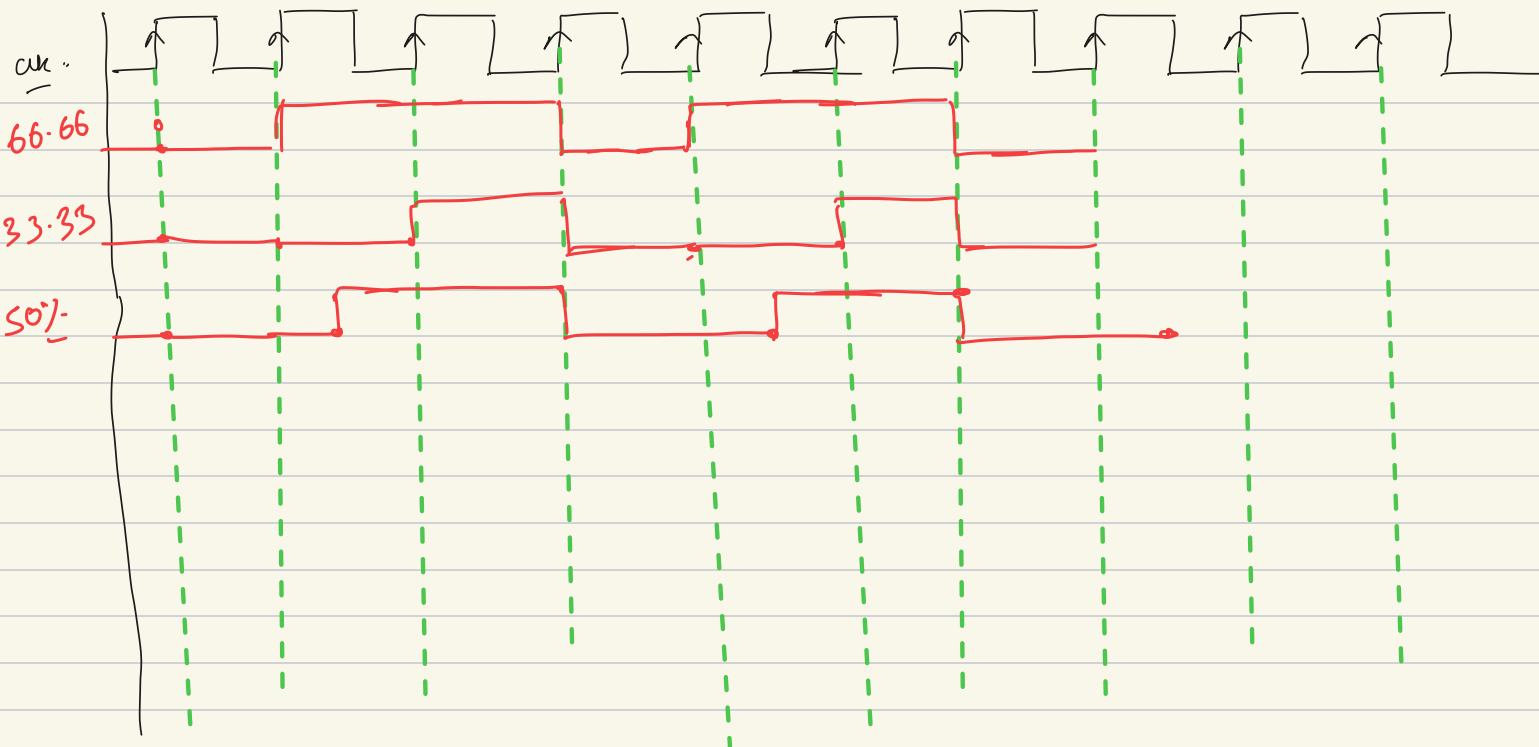
$$D = \frac{2}{10} = 0.2 = 20\%$$

So the duty cycle is **20%**. NI

by odd no's.

6/3 with 33.33 percent
50 percent
66.66 percent duty -

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$$\frac{2}{2+1} = \frac{2}{3} = 66.66\%$$

How To design?

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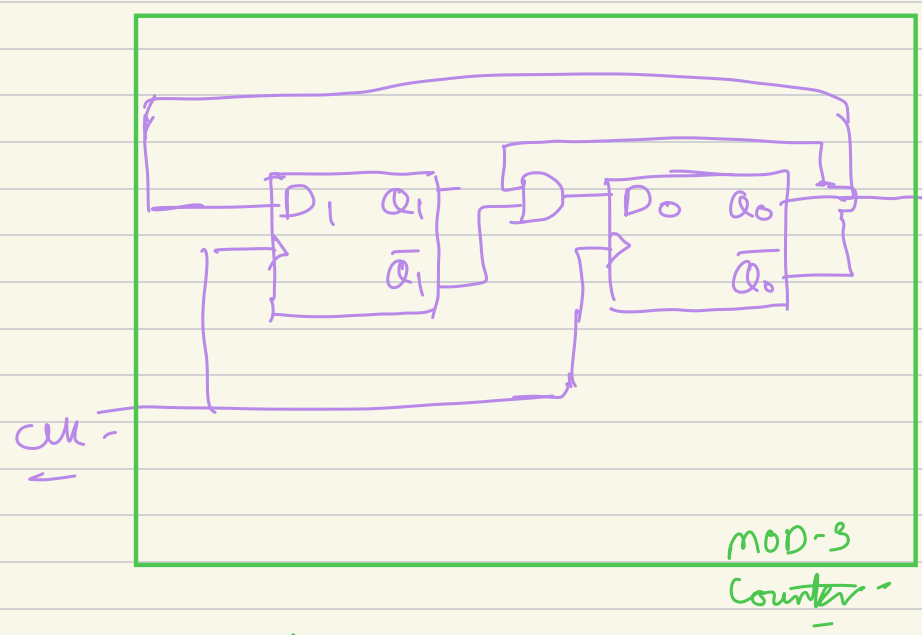
MOD 3 Counter

Q_1	Q_0
0	0
0	1
1	0

Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0
0	0	0	1	0	1
0	1	1	0	1	0
1	0	0	0	0	0

$D_1 = \overline{Q_1} Q_0 = Q_0$

$D_0 = \overline{Q_1} \overline{Q_0}$

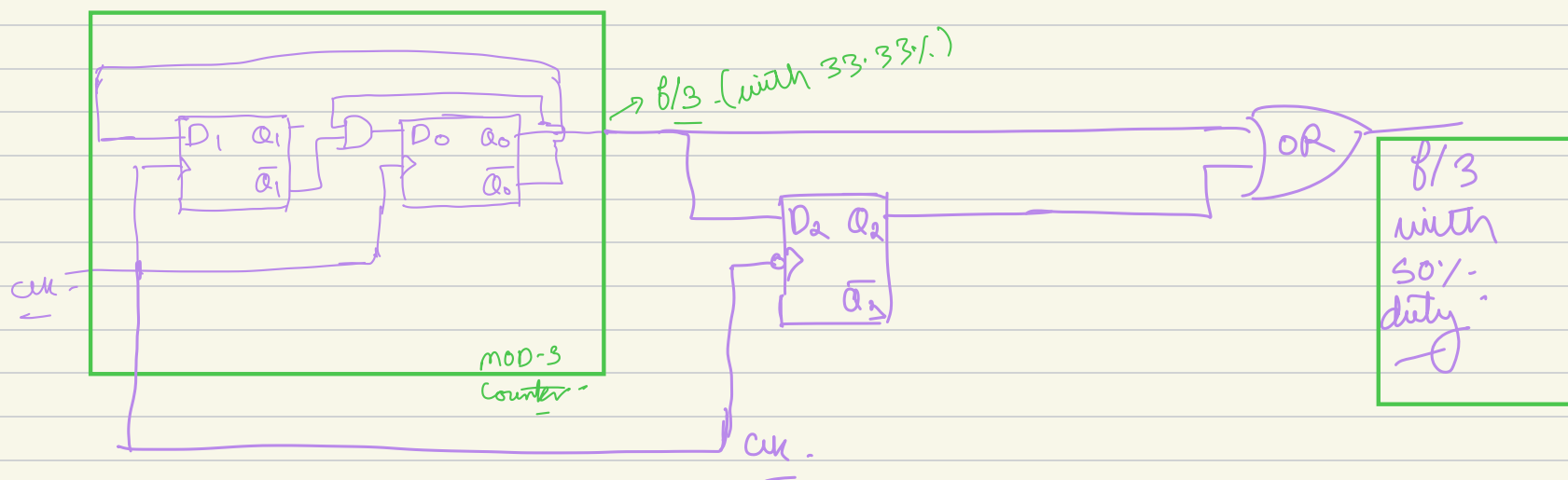


→ 1/3 (with 33.33%)

MOD-3 Counter

Q_1	Q_0
0	0
0	1
1	0

33.33%



for 50% duty cycle.

8/5 ckt.

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Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	0	0	0	0	0	0

$D_2 = \bar{Q}_1 \bar{Q}_0 \bar{Q}_1 Q_0 Q_1 Q_0 Q_1 \bar{Q}_0$

$\bar{Q}_2$	0	0	1	0
Q_2	0	X	X	X

$= Q_1 Q_0 = D_2$

$\bar{Q}_1$

Q_0

Q_2	Q_1	Q_0
$\frac{1}{5}$	$\frac{2}{5}$	$\frac{2}{5}$
$= 20\%$	$\downarrow 40$	$\downarrow 40$

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D_0

$\bar{Q}_1 \bar{Q}_0 \bar{Q}_1 Q_0 Q_1 Q_0 Q_1 \bar{Q}_0$

$\bar{Q}_2$	1	0	0	1
Q_2	0	X	X	X

$D_0 = \bar{Q}_2 \bar{Q}_0$

D_1

$\bar{Q}_1 \bar{Q}_0 \bar{Q}_1 Q_0 Q_1 Q_0 Q_1 \bar{Q}_0$

$\bar{Q}_2$	0	1	0	1
Q_2	0	X	X	X

$D_1 = \bar{Q}_1 Q_0 + Q_1 \bar{Q}_0 = Q_1 \oplus Q_0 \sim$

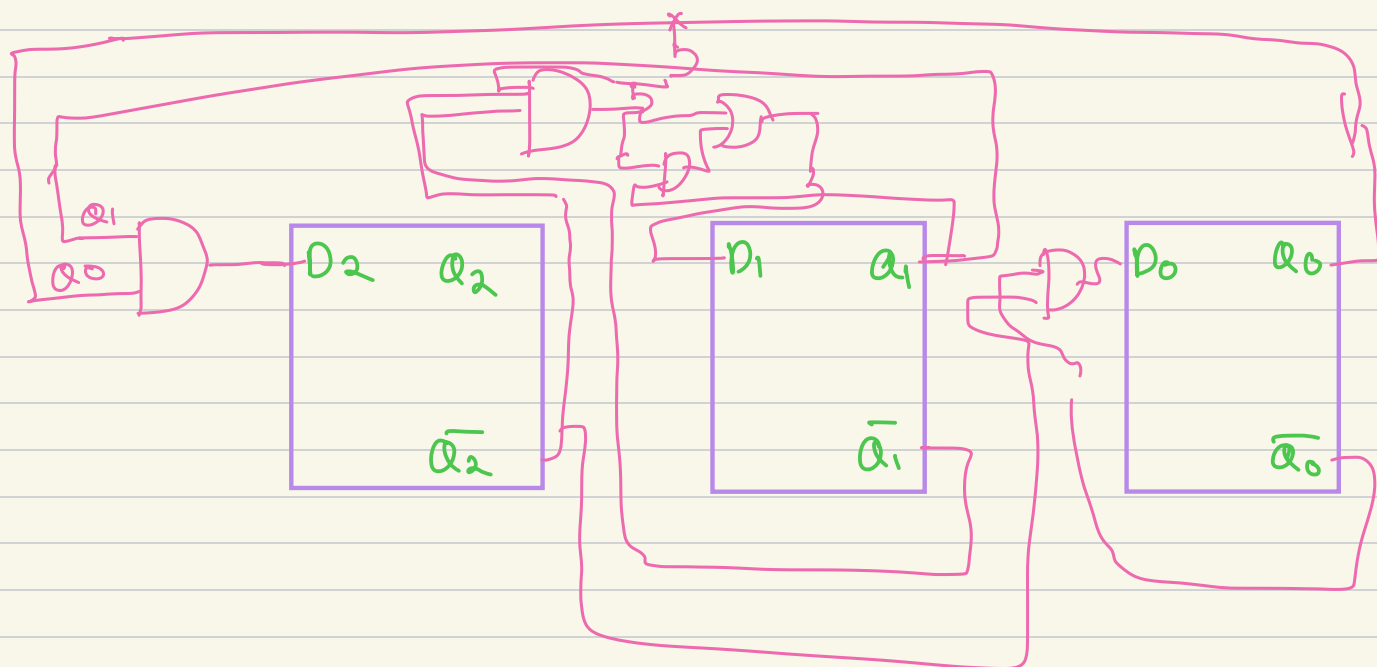
$Q_1 Q_0 = D_2$

$D_0 = \bar{Q}_2 \bar{Q}_0$

$$D_1 = \overline{Q_2} \overline{Q_1} Q_0 + Q_1 Q_0$$

$$Q_1 Q_0 = D_2$$

$$D_0 = \overline{Q_2} \overline{Q_0}$$



Statement

Q_2	Q_1	Q_0
20%	40%	40%
$(1/5)$	$(2/5)$	$(2/5)$
$\overline{Q_2}$	$\downarrow$	$\overline{Q_0}$
80%	= 60%	60%

$Q_1 = 40\%$ duty

If I want 50%

Q_2 + Negative triggered

OR Q_2

Q_2	Q_1	Q_0
0	0	1
0	1	0
0	1	0
1	0	0
1	0	1

Duty =

$$\frac{2}{5} = 40\%$$

3/5
60%

2/5

40

40

60

40

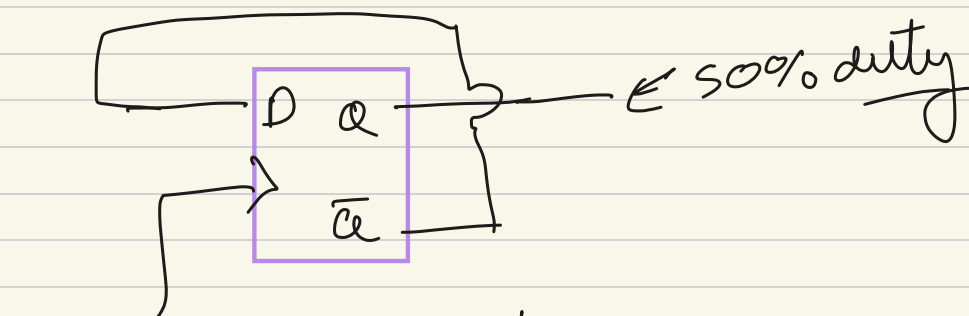
Digi Qs Solution

Q10-

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Q10) Design a frequency divide-by-2 circuit using DFF and external gates which gives
(a) 50% duty cycle (b) 25% duty cycle?

Divide by 2

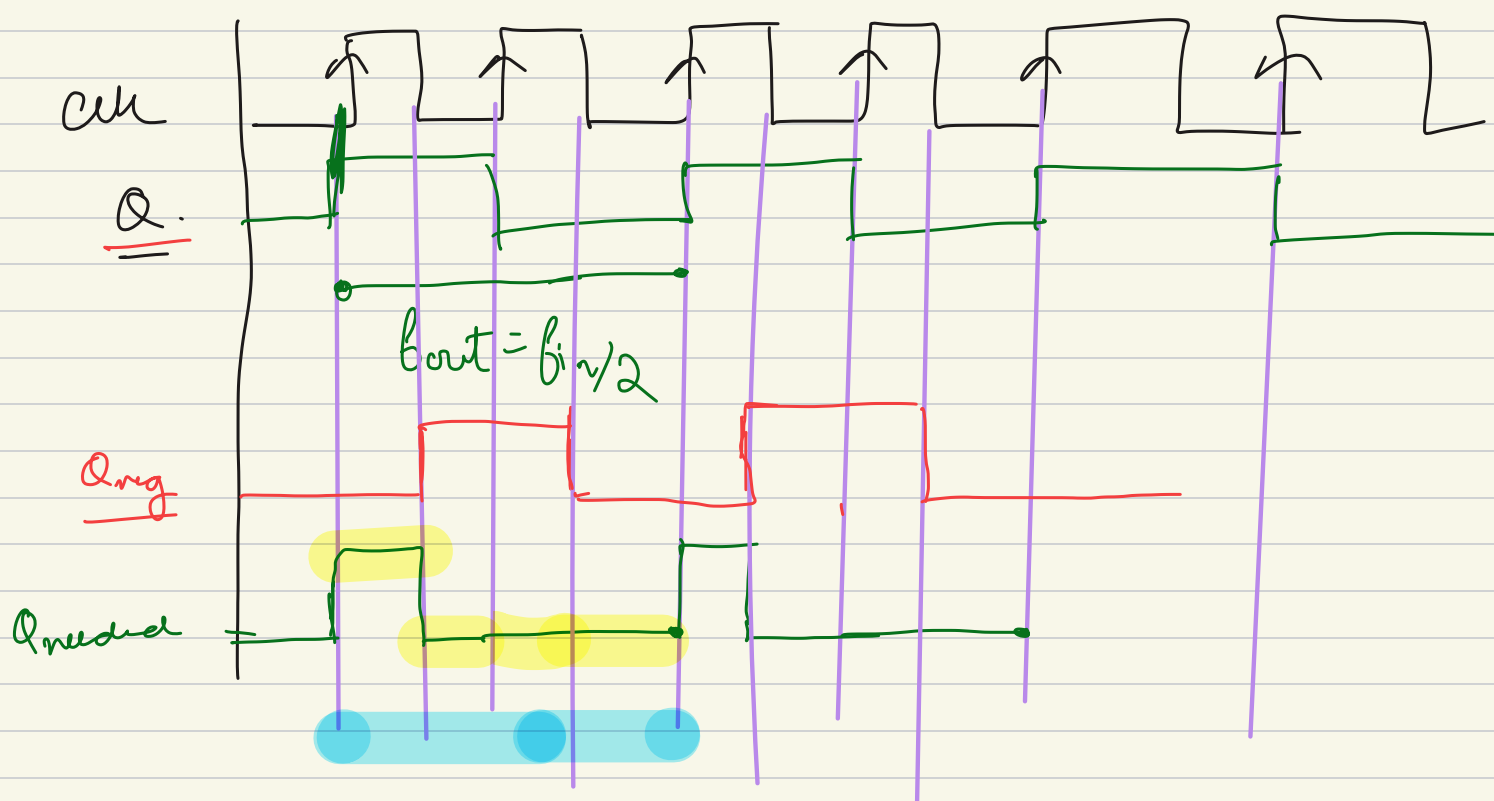


Q_0	Q_0^+	D
0	1	1
1	0	0

$$D = \overline{Q_0}$$

b) Now To make 25% duty - ?

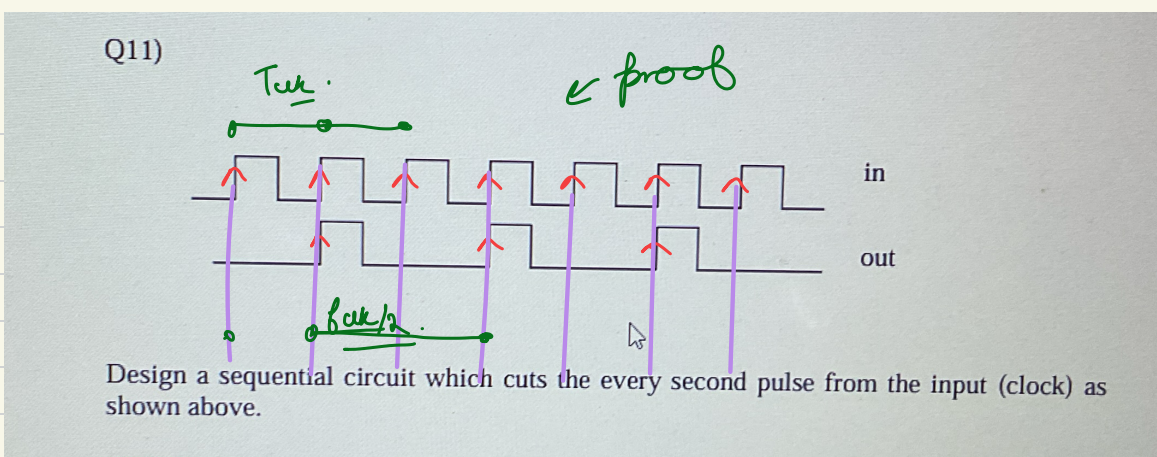
True



Anding Q & $\overline{Q}$ through a neg triggered is giving 25% duty

OR
ANDing b_{in} & b_{out} True

Q11



$buck/2$ with 25% duty cycle. ✓

Q12. a) → $6/3$ with $\frac{1}{3}$ or 33.33% duty.

Q_1	Q_0	Q_1^+	Q_0^+	D_1	D_0
0	0	0	1	0	1
0	1	1	0	1	0
1	0	0	0	0	0

$D_1 = \overline{Q_0} Q_0$

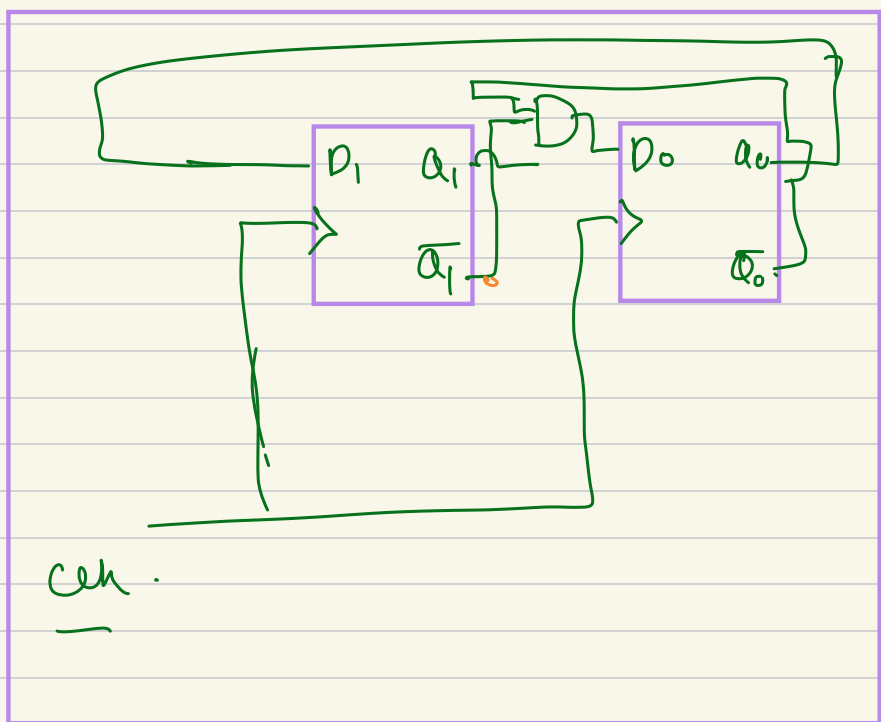
Q_1	$\overline{Q_0}$	Q_0
0	1	0
1	0	1

$D_0 = \overline{Q_1} \overline{Q_0}$

$\overline{Q_1}$	$\overline{Q_0}$
1	1
0	0

$D_1 = Q_0$

$D_0 = \overline{Q_1} \overline{Q_0}$



Q_1	Q_0
0	0
0	1
1	0

Q_1 & Q_0 are $6/3$ (33.33%) ✓

b)

Q12 -

Q13 → a) To get a) 66.67% .

b) 50% duty.

Q_1	Q_0
0	0
0	1
1	0

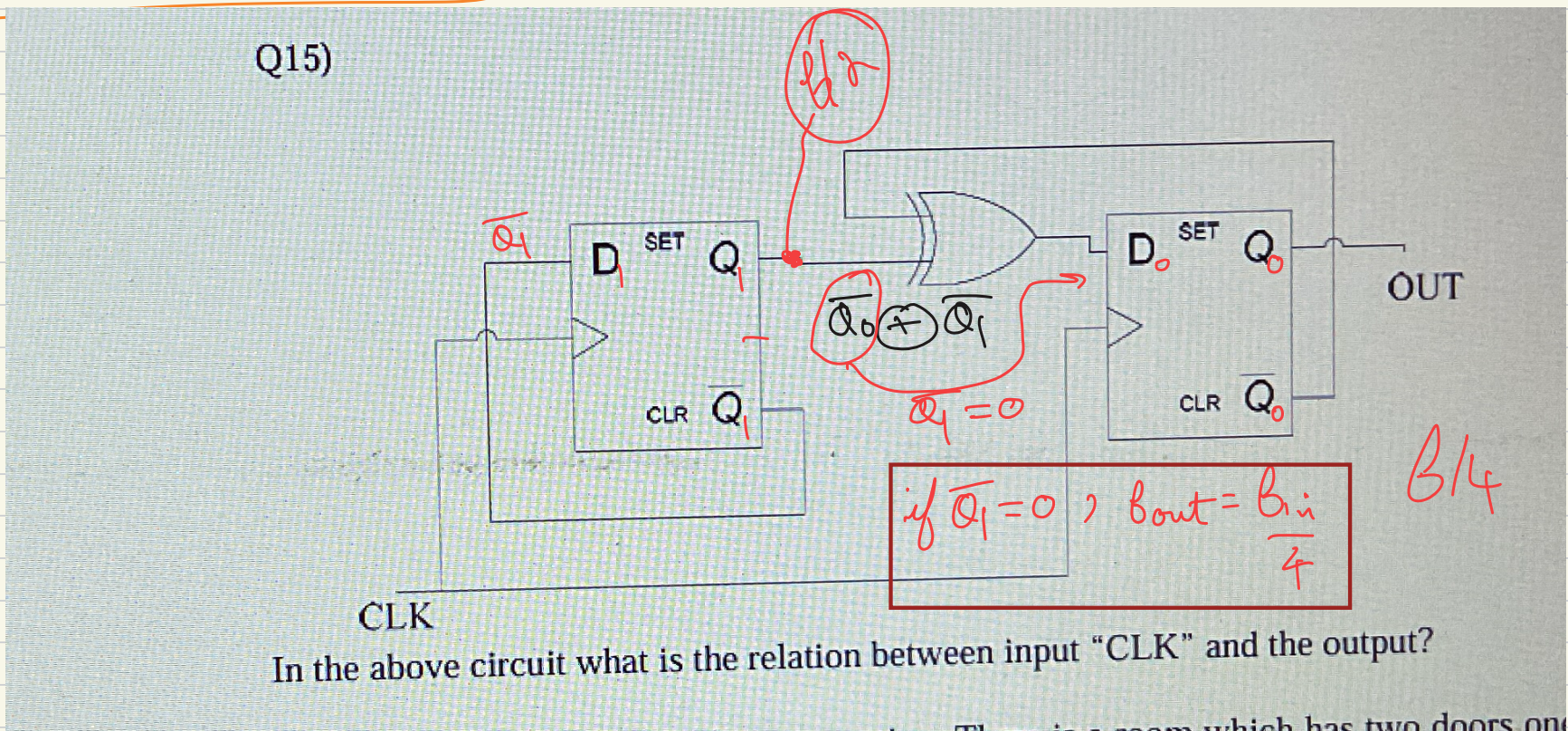
cek .

66.66

66.66

B/3 @ 50%

$b_{out} = 2/3 b_{in}$



Q_1	Q_0	Q_1^+	Q_0^+
0	0	1	
0	1	1	
1	0	0	
1	1	0	

	Q_1	Q_0
$q_1 = q_0 = 0$	0	0
$q_1 = 1$	0	1
$q_1 = 0$	1	0
$q_1 = 1$	1	1

Handwritten notes: $B/4$ (circled), $1/2$ (written next to the first row), and a large bracket on the right side of the table.

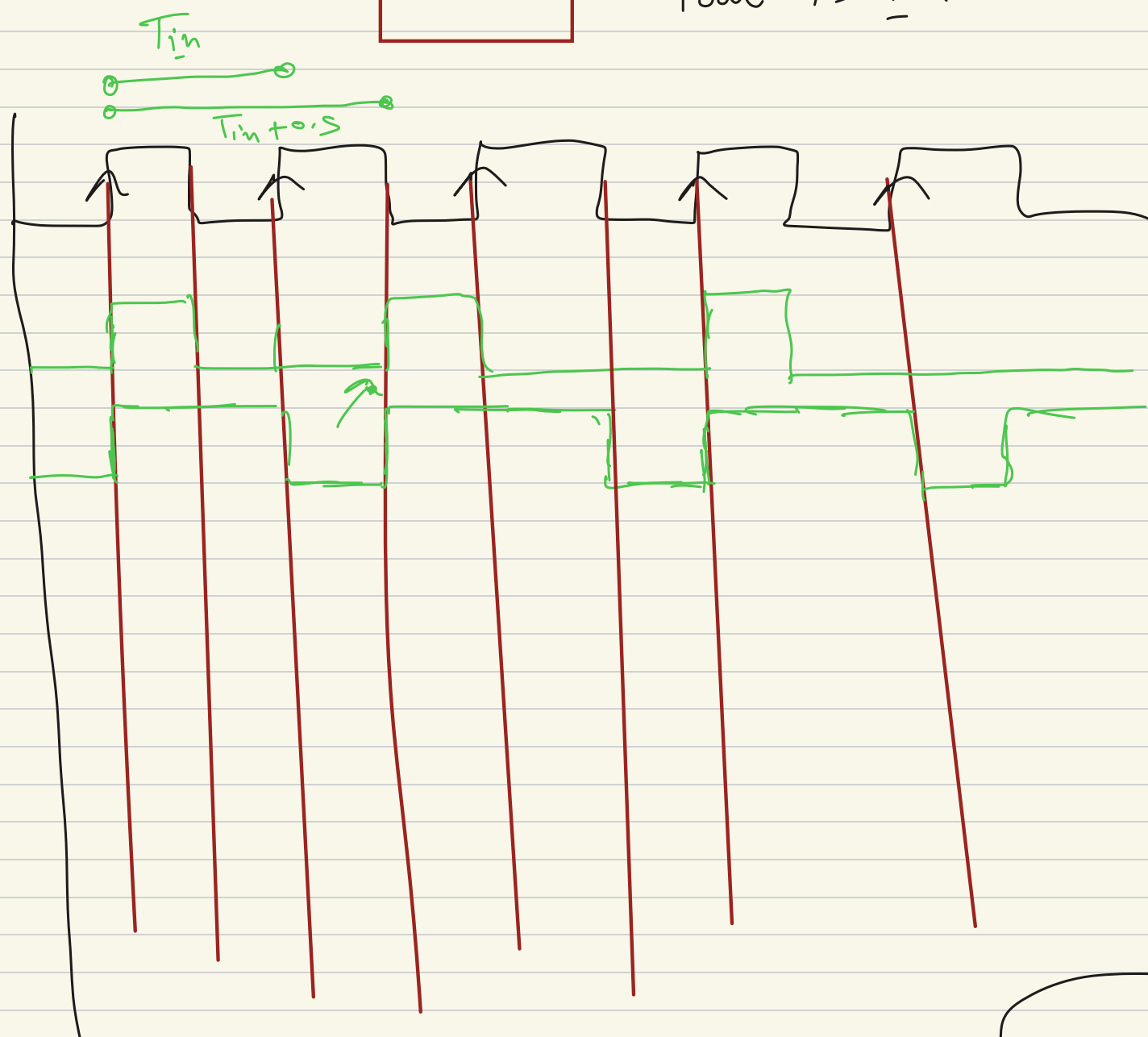
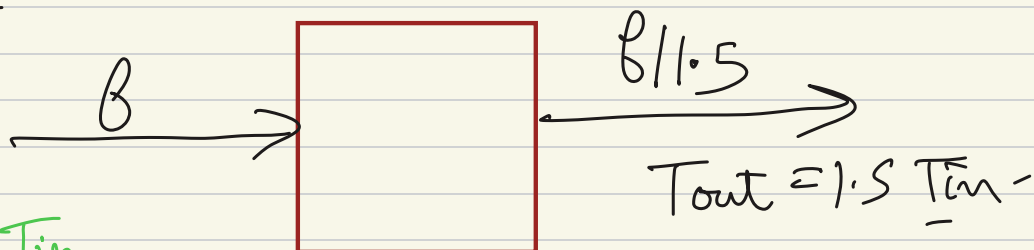
divider of $b_{out} = (2/3) b_{in}$

$Q_1^+ = \overline{Q_1}$ Toggle Mode.

$$b_{out} = \frac{2}{3} b_{in}$$

$$\frac{b_{out}}{b_{in}} = \frac{2}{3} \Rightarrow b_{out} = \frac{2}{3} b_{in} = \frac{b_{in}}{\left(\frac{3}{2}\right)} = \frac{b_{in}}{1.5}$$

$N=105$ —



These are same b_{out}
 $= \frac{b_{in}}{1.5}$
just different duty.

$$b_{out} = \frac{b_{in}}{1.5}$$

$$\frac{1}{T_{out}} = \frac{1}{1.5} \frac{1}{T_{in}} = T_{out} = 1.5 T_{in}$$

2 blocks.

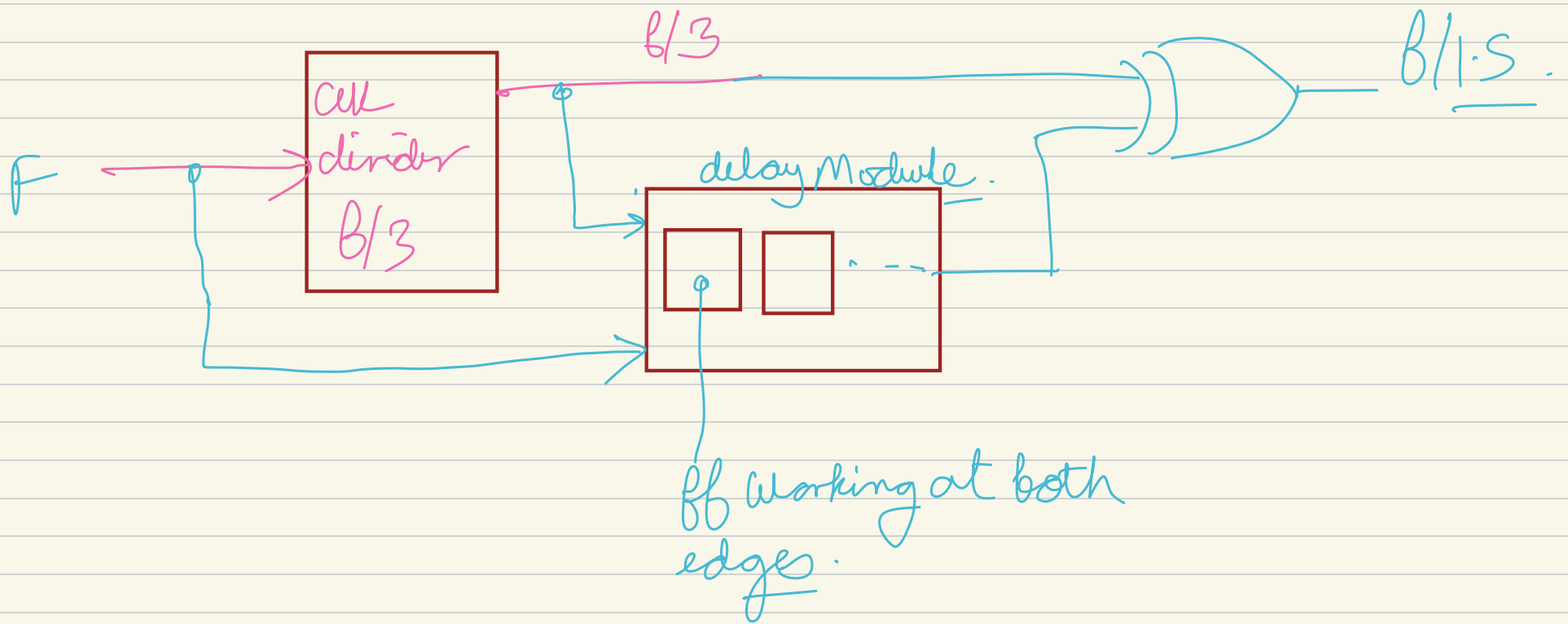
①

F/n divider
here n is a
integer.

BB, that
works on
both edges.

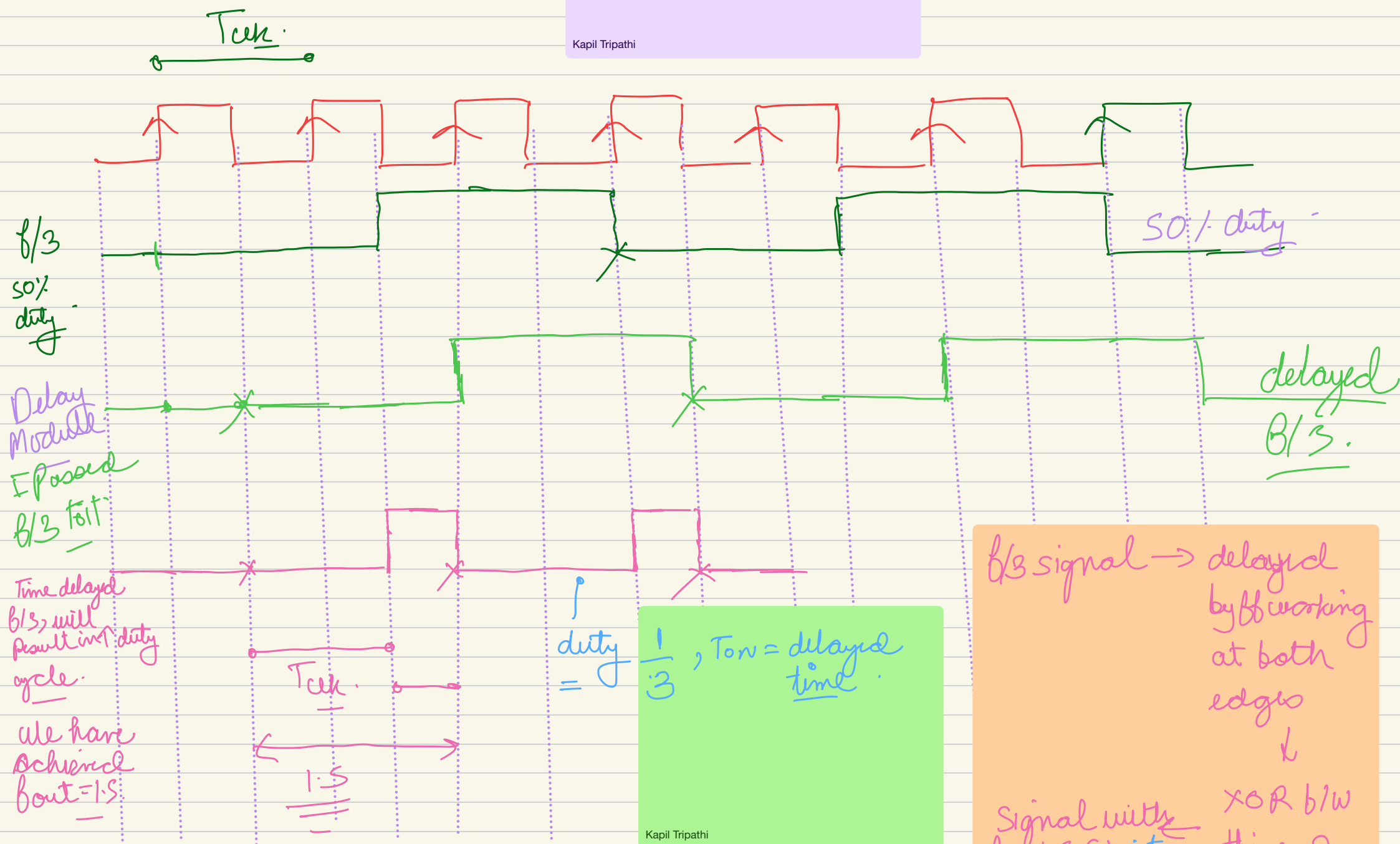
Must have 50% duty.

$f/1.5 \rightarrow$ step 1 $\rightarrow 1.5 \times 2 = (3)$
 so $f/3$



This delay Module will tell the duty cycle.

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$f/3$ signal $\rightarrow$ delayed by fb working at both edges

$\downarrow$

Signal with $f = 1.5$ (i.e. its repeating 2 times)

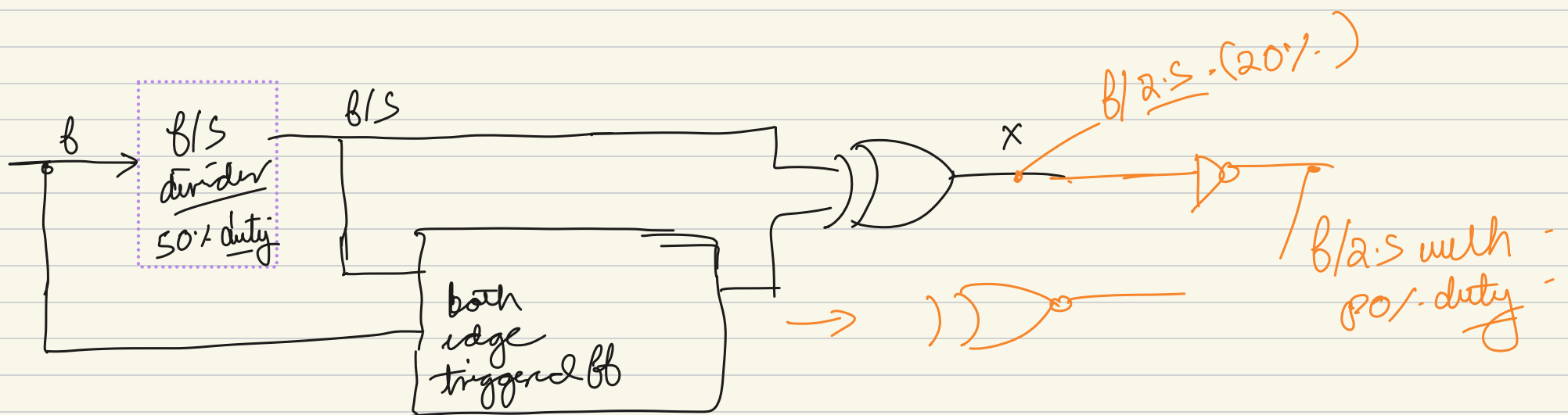
XOR b/w these 2

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- ① Clock freq divider
+ Delay Module
+ XOR gate
- ② Multiply n by 2
($b/2n$) cause $2n$ is an integer number -
 $(1.5 \times 2) = b/3$.

② b/n must have a duty cycle of 50%.

Q. $b/2.5 \rightarrow b/2.5 \times 2 = b/5$.



$b/5 + 1$ bb in delayed will give a pulse, like
of only time period $T_{clk}/5$
duty = $\frac{1}{5} = 20\%$ ✓

Cause I am just using $b/5$ and 1 edge delayed
 $b/5$.

2 bb will give me 40% duty cycle.

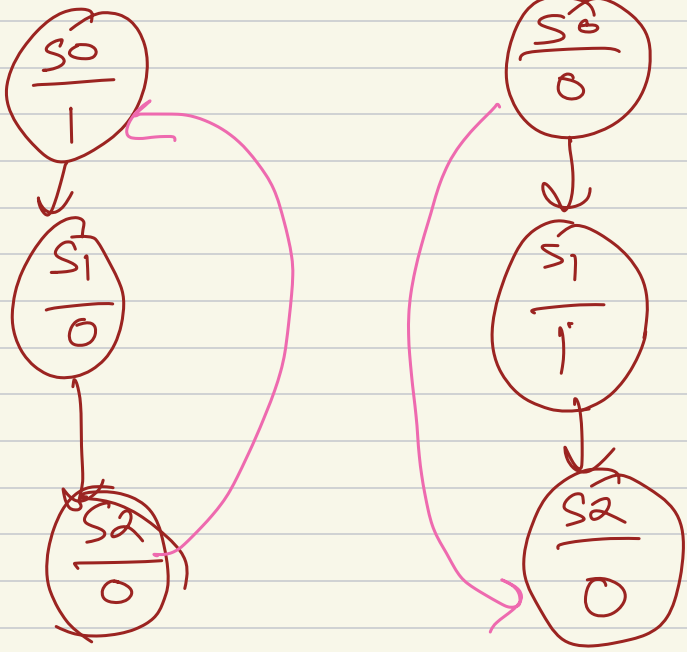
Duty cycle of 20% then , $b/2.5$ with 80% saving upon 3 re-edge triggered bb

Q. clock frequency divider
 $N=1.5$ by FSM

POS-sm.

Neg edge

$1.5 \times 2 = 3$ clock cycles of
 I/P clock.



ORing

RTL + design

$$\frac{b_{out}}{b_{in}} = \frac{2}{3} \Rightarrow b_{out} = b_{in} \cdot \frac{2}{3} = \left(\frac{b_{in}}{1.5} \right)$$

$B/3$ with 50% $\rightarrow B/3$ & the neg edge triggered